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```
clc;
clear all;
close all;
```

Inputs, Variables, Constants

```
alphabets = ["A", "B", "C" , "D"];
probabs = [0.3 0.3 0.2 0.2];
MESSAGE = 'ABBACD';
```

Interactive Inputs

```
%alphabets = input("Enter alphabets array in ascending order: ");
%probabs = input("Enter respectively probability array: ");
%MESSAGE = input("Enter message: ");
```

Algorithm

```
cumProbs = [0 cumsum(probabs)];

low = 0;
high = 1;

fprintf('Initial range: [%.6f , %.6f)\n\n', low, high);

for k = 1:length(MESSAGE)
    idx = find(alphabets == MESSAGE(k));

    % Current range width
    range = high - low;

    % Calculate new boundaries CORRECTLY
    low_new = low + range * cumProbs(idx);
    high_new = low + range * (cumProbs(idx) + probabs(idx));

    % Update for next iteration
    low = low_new;
    high = high_new;

    fprintf('After %c : [%.6f , %.6f)\n', MESSAGE(k), low, high);
end
```

```
Initial range: [0.000000 , 1.000000)

After A : [0.000000 , 0.300000)
After B : [0.090000 , 0.180000)
After B : [0.117000 , 0.144000)
After A : [0.117000 , 0.125100)
After C : [0.121860 , 0.123480)
After D : [0.123156 , 0.123480)
```

Results

```
fprintf('\nEncoded range for the message "%s" is: [%f , %f)\n',
 MESSAGE, low, high);
res = (low + high) / 2;
fprintf('Assigned average value for the message is: %f\n', res);
```

```
Encoded range for the message "ABBACD" is: [0.123156 , 0.123480)
Assigned average value for the message is: 0.123318
```

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